0152. Maximum Product Subarray

Medium · 1-D Dynamic Programming
leetcode.com/problems/maximum-product-subarray

Problem

- Return the largest product of a contiguous non-empty subarray.

Intuition

- A negative number flips the biggest product into the smallest and vice versa.
- Track both the max and the min product ending at each index.

Brute Force

- All subarrays: O(n^2).

Optimal Approach

- Hi = lo = best = nums[0].
- For x: candidates x, hix, lox; hi = max, lo = min; best = max(best, hi).

Worked Example

```
[2,3,-2,4].

x=2  hi=2  lo=2   best=2
x=3  hi=6  lo=3   best=6
x=-2 hi=-2 lo=-12 best=6
x=4  hi=4  lo=-48 best=6
```

Java Solution

```java
public class MaximumProductSubarray {
    public int maxProduct(int[] nums) {
        int hi = nums[0], lo = nums[0], best = nums[0];
        for (int i = 1; i < nums.length; i++) {
            int x = nums[i], a = hi * x, b = lo * x;
            hi = Math.max(x, Math.max(a, b));
            lo = Math.min(x, Math.min(a, b));
            best = Math.max(best, hi);
        }
        return best;
    }
}
```

Complexity

- Time: O(n). Space: O(1).

Pitfalls

- Compute the new hi and lo from the old values simultaneously (use temps).
- Zero resets both, which the "x alone" candidate handles.